Week 8
Deep Learning (IIT Ropar)

Data for questions from 1 to 6

Employee Attrition Prediction in a Public Sector IT Organization

A large public sector IT organization in India employs thousands of software engineers and support staff across multiple locations. In recent years, the organization has observed an increase in employee attrition, especially among early-career professionals. High attrition leads to increased recruitment costs, loss of trained manpower, and disruption of long-term projects.

To better understand and predict attrition trends, the organization's analytics division decides to build a deep learning–based employee attrition prediction model. The objective is to predict whether an employee is likely to leave the organization within the next six months. The dataset consists of 12,000 employee records, each described using numerical features such as:

- Years of experience
- Average monthly working hours
- Performance rating (scaled between 0 and 1)
- Number of projects handled
- Training hours completed in the last year

The output is a binary label:

- 1 → Employee likely to leave
- 0 → Employee likely to stay

The data is split into 80% training data and 20% validation data. A deep feedforward neural network with multiple hidden layers is designed for this task.

Initially, the data science team uses the sigmoid activation function in all hidden layers. During training, they observe that although the loss decreases initially, learning slows down significantly as the network depth increases. The gradients in earlier layers become extremely small, making weight updates negligible. As a result, the training process becomes inefficient.

To address this issue, the team replaces the sigmoid activation in hidden layers with the ReLU activation function, while keeping the output layer unchanged. After this change, the network begins to train faster, and the loss decreases more consistently across epochs.

For analysis purposes, consider a neuron in a hidden layer receiving an input value of **–2** and **+3** during training. The team computes the activation output for both sigmoid and ReLU functions to understand their behavior.

The sigmoid activation function is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The ReLU activation function is defined as:

$$f(x) = \max(0, x)$$

By comparing these activation values and understanding how gradients behave for different activation functions, the team evaluates why ReLU is more suitable for training deep networks in this employee attrition prediction problem.

Question: 01

Why does using sigmoid activation in deep hidden layers slow down training?
Options:

A) Sigmoid outputs are always zero
B) Sigmoid causes exploding gradients
C) Sigmoid leads to vanishing gradients
D) Sigmoid increases model variance

Correct Answer: C

Explanation: Sigmoid squashes inputs into a small range, causing gradients to become very small in deeper layers.

Question: 02

Which property of ReLU helps improve training speed in deep neural networks?

Options:

A) Outputs are bounded between 0 and 1
B) It introduces non-linearity by squaring inputs
C) It avoids gradient saturation for positive inputs
D) It normalizes the input distribution

Correct Answer: C

Explanation: ReLU does not saturate for positive inputs, allowing gradients to flow effectively.

Question: 03

What is the approximate output of the **sigmoid activation** for an input value of **+3**?

Options:

A) 0.05
B) 0.50
C) 0.95
D) 3.00

Correct Answer: C

Question: 04

Which of the following statements about activation functions are correct in this case?

Options:

A) Sigmoid activation can cause vanishing gradients in deep networks
B) ReLU outputs zero for negative inputs
C) ReLU always produces outputs between 0 and 1
D) ReLU improves gradient flow for positive inputs

Correct Answers: A, B, D

Explanation: ReLU outputs zero for negative inputs and allows better gradient flow for positive inputs, while sigmoid suffers from vanishing gradients.

Question: 05

Why is ReLU preferred over sigmoid in the hidden layers for this employee attrition model?

Options:

A) Faster convergence during training
B) Reduced risk of vanishing gradients
C) Guaranteed higher validation accuracy
D) Better scalability to deeper networks

Correct Answers: A, B, D

Explanation: ReLU improves optimization efficiency and enables training of deeper models.

Question: 06

Fill in the blank

If the input to a ReLU neuron is **-3**, the output of the neuron is _____.

Correct Answer: 0

Data for questions from 7 to 11

Traffic Flow Prediction for an Indian Smart City

As part of the Smart Cities Mission, the municipal corporation of a large Indian metropolitan city deploys an intelligent traffic management system to reduce congestion during peak hours. The city collects traffic data from sensors installed at major junctions, flyovers, and arterial roads. These sensors continuously record traffic-related statistics that can help predict congestion levels in advance.

The collected dataset contains hourly traffic records from multiple junctions over several years. Each data sample includes numerical features such as:

- Number of vehicles passing through the junction
- Average vehicle speed (km/h)
- Traffic signal waiting time (seconds)
- Road occupancy percentage
- Time of day encoded numerically

The goal is to predict traffic congestion level, represented as a continuous numerical value. To model this complex relationship, the data science team builds a deep feedforward neural network with eight hidden layers, each consisting of 128 neurons, and uses ReLU activation functions in all hidden layers.

Initially, the team initializes all network weights using a naïve random initialization with variance equal to 1. During training, they observe the following issues:

- Activations in deeper layers become excessively large
- Gradients fluctuate significantly during backpropagation
- Training loss oscillates instead of decreasing smoothly

To understand this behavior, the team analyzes how variance propagates through layers in deep networks. They find that when weights are not properly initialized, the variance of activations can grow exponentially with depth, especially in networks using ReLU activations. To stabilize training, the team switches to He initialization, which is specifically designed for networks with ReLU activations. He initialization sets the variance of the weights according to:

$$Var(w) = \frac{2}{n}$$

where n is the number of input connections to a neuron.

For comparison, the team also experiments with Xavier initialization, where the variance is given by:

$$Var(w) = \frac{1}{n}$$

After switching to He initialization, the team observes that:

- Activation variances remain stable across layers
- Gradients neither vanish nor explode
- Training loss decreases smoothly and consistently

To quantify this improvement, consider a hidden layer receiving inputs from 128 neurons. The team computes the expected variance of the initialized weights using both Xavier and He initialization strategies and compares their impact on training stability.

The final objective of the city administration is to deploy a model that is numerically stable, efficient to train, and reliable, especially during high-traffic events such as festivals and public gatherings.

Question: 07

Why did naïve random initialization lead to unstable training in this deep traffic prediction network?

Options:

A) The dataset was too small
B) Variance of activations increased across layers
C) ReLU activation removes non-linearity
D) Learning rate was fixed

Correct Answer: B

Explanation: Improper initialization causes activation variance to explode or vanish as depth increases.

Question: 08

Why is He initialization preferred over Xavier initialization for this network?

Options:

A) The network uses sigmoid activations
B) The network uses ReLU activations
C) He initialization reduces model depth
D) Xavier initialization increases variance

Correct Answer: B

Explanation: He initialization accounts for the fact that ReLU outputs zero for roughly half of its inputs.

Question: 09

If a hidden layer has 128 input neurons, what is the variance of weights using Xavier initialization?

Options:

A) 0.0078
B) 0.0156
C) 0.0312
D) 0.0625

Correct Answer: A

Question: 10

Which of the following outcomes indicate successful training stabilization after switching to He initialization?

Options:

A) Stable activation variance across layers
B) Smooth decrease in training loss
C) Guaranteed zero test error
D) Reduced risk of exploding gradients

Correct Answers: A, B, D

Explanation: Proper initialization stabilizes forward and backward signal propagation but does not guarantee perfect accuracy.

Question: 11

Why is correct initialization especially critical in deep networks like this traffic prediction model?

Options:

A) Errors amplify across multiple layers
B) Initialization determines gradient magnitude
C) Poor initialization slows convergence
D) Initialization eliminates the need for normalization

Correct Answers: A, B, C

Explanation: Depth amplifies initialization effects, impacting both learning speed and stability.

Data for questions from 12 to 15

Speech Command Recognition System in Indian Railways

Indian Railways is modernizing station operations by introducing voice-enabled kiosks and hands-free control panels for announcements, ticketing assistance, and staff coordination. These systems must reliably recognize short spoken commands (e.g., "platform change," "emergency stop," "signal check") in noisy environments such as platforms and control rooms.

To build a robust solution, the engineering team develops a deep neural network–based speech command recognition system. The input to the model consists of numerical acoustic features extracted from audio signals, such as Mel-frequency cepstral coefficients (MFCCs), energy, and spectral features. Each audio sample is represented as a fixed-length numeric vector.

The dataset contains 40,000 labeled audio samples collected from multiple railway stations across India, recorded under varying noise conditions (crowds, announcements, train engines). The target output is a multi-class label, representing one of several predefined commands.

The team initially trains a deep feedforward neural network with 10 hidden layers, each containing 256 neurons, using ReLU activation functions. Despite careful tuning of the learning rate and proper weight initialization, they observe the following problems during end-to-end supervised training:

- Training loss decreases very slowly
- Gradients in the lower (earlier) layers become extremely small
- Deeper layers learn meaningful representations, but earlier layers remain poorly trained

These symptoms indicate difficulty in training very deep networks, even when modern activation functions and initialization strategies are used.

To overcome this issue, the team adopts Greedy Layer-wise Pretraining, a technique discussed in Lecture 9 for training deep architectures. Instead of training the entire network end-to-end from scratch, the network is trained one layer at a time in a greedy manner.
The procedure followed is:

1. The first hidden layer is trained using the input features and a simple objective.
2. Once trained, the parameters of the first layer are frozen.
3. The output of the first layer becomes the input to the second layer.
4. The second layer is trained while keeping earlier layers fixed.
5. This process continues layer by layer until all hidden layers are pretrained.
6. Finally, the entire network is fine-tuned end-to-end using supervised learning.

During pretraining analysis, consider a particular hidden layer that receives 128 input features. The layer uses ReLU activation and He initialization. The variance of weights is set accordingly to maintain stable signal propagation.

After applying greedy layer-wise pretraining followed by fine-tuning, the team observes:

- Faster convergence during supervised training
- Improved gradient flow to earlier layers
- Higher validation accuracy compared to training without pretraining

The railway authorities must decide whether this approach is suitable for large-scale deployment, especially given the safety-critical nature of voice-controlled railway systems.

Question: 12

What is the primary motivation for using greedy layer-wise pretraining in this deep speech recognition model?

Options:

A) To increase dataset size
B) To simplify the network architecture
C) To improve training of earlier layers
D) To eliminate the need for fine-tuning

Correct Answer: C

Explanation: Greedy pretraining helps earlier layers learn meaningful representations before end-to-end training.

Question: 13

During greedy layer-wise pretraining, what happens to the parameters of previously trained layers?

Options:

A) They are randomly reinitialized
B) They are frozen
C) They are removed from the network
D) They are updated with a higher learning rate

Correct Answer: B

Explanation: Previously trained layers are frozen while training the next layer.

Question: 14

Why does greedy layer-wise pretraining help alleviate vanishing gradient issues?

Options:

A) It removes nonlinear activation functions
B) It ensures each layer learns useful representations independently
C) It increases the learning rate automatically
D) It reduces the depth of the network

Correct Answer: B

Explanation: Each layer is trained to produce meaningful outputs before gradients must pass through many layers.

Question: 15

Which of the following are valid benefits of greedy layer-wise pretraining in this case?

Options:

A) Faster convergence during supervised fine-tuning
B) Improved gradient flow in early layers
C) Guaranteed perfect recognition accuracy
D) Better initialization for deep networks

Correct Answers: A, B, D

Explanation: Pretraining improves optimization and initialization but does not guarantee perfect accuracy.